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Hand-held Object Reconstruction from RGB Video with Dynamic Interaction

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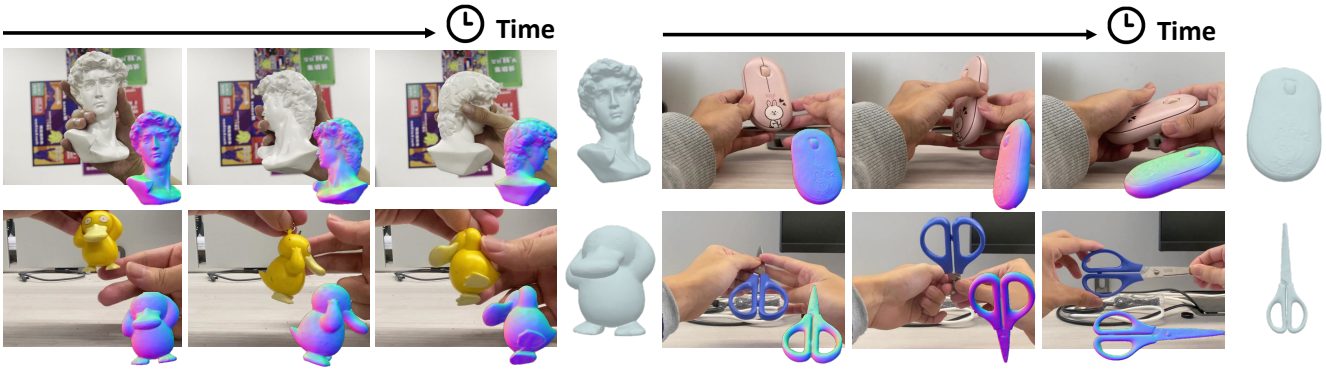


Figure 1. **Hand-held object reconstruction.** Given a monocular RGB sequence of one or both hands manipulating a rigid object, our method accurately reconstructs the object even when it is poorly textured. Here, we present reconstructed surface normals and meshes from both lab-recorded and self-captured mobile phone videos.

Abstract

This work aims to reconstruct the 3D geometry of a rigid object manipulated by one or both hands using monocular RGB video. Previous methods rely on Structure-from-Motion or hand priors to estimate relative motion between the object and camera, which typically assume textured objects or single-hand interactions. To accurately recover object geometry in dynamic interactions, we incorporate priors from 3D generation model into object pose estimation and propose semantic consistency constraints to solve the challenge of shape and texture discrepancy between the generated priors and observations. The poses are initialized, followed by joint optimization of the object poses and implicit neural representation. During optimization, a novel pose outlier voting strategy with inter-view consistency is proposed to correct large pose errors. Experiments on three datasets demonstrate that our method significantly outperforms the state-of-the-art in reconstruction quality for both single- and two-hand scenarios. Our project page: https://east-j.github.io/dynhor/

1. Introduction

Modeling the 3D shapes of objects has numerous applications in fields such as virtual/augmented reality and robotic manipulation. Compared to the common approach of capturing static objects by manually moving a camera around them, rotating the object by hand in front of a camera or head-mounted device offers greater convenience for applications involving dynamic hand-object interactions. This approach also enables reconstruction of the entire object surface, including the bottom part. This hand-held object reconstruction problem is the focus of the work.

Recent hand-held object reconstruction methods generally adopt multi-view reconstruction techniques [49, 54] to aggregate observations of in-hand objects across multiple rigid poses. The key challenge lies in accurately estimating the relative motion between the object and the camera. Some works [18, 21, 42] assume the object is firmly grasped, tracking object motion with a hand pose tracker instead. This constrains hand-object movement and causes parts of the object to be consistently occluded by the hand, requiring post-processing or prior knowledge to obtain a complete surface, which limits reconstruction quality and the potential application scenarios. Another line of method [9] uses Structure-from-Motion (SfM) to initialize object poses and refine them with hand priors. However, as pointed out by [13, 43], SfM struggles in dynamic scenes. The solution is also limited to single-hand interactions due to using hand priors. To overcome these limitations, Hampali et al. [13] proposed an incremental training approach that divides sequences into trackable segments for joint shape and pose optimization. While effective for two-hand interactions, this approach struggles with texture-less

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or thin-structured objects and suffers from pose errors at segment boundaries, which degrade reconstruction quality.

Towards enabling accurate reconstruction of a rigid object manipulated dynamically by one or both hands, we propose a novel hand-held object reconstruction pipeline consisting of pose initialization with generated 3D object priors and joint optimization for object poses and reconstruction with pose outlier voting. For methods using hand priors, the high-DOF deformable hand itself is hard to resolve and not robust in challenging environments. Additionally, occlusions and dynamic objects make SfM-based methods brittle. Instead, recent 3D object generation models can compensate for occluded information for hand-held object reconstruction and eliminate the unstable high-DOF deformable part from the algorithm.

For pose initialization, our approach generates object prior from the pre-trained 3D generation model to constrain the pose space, eliminating the need for SfM or hand priors. One major challenge of utilizing the generated 3D models is the shape and texture discrepancy from the ground truth: the discrepancy can cause registration methods that minimize the appearance difference between a frame and the rendered counterpart from the model to fail easily. We propose using semantic consistency to address the appearance discrepancy in shape and texture for pose initialization with generated 3D models, *e.g*. the distance between caps of bottles in different sizes, shapes, and textures in the semantic manifold should be smaller than the distance between the caps and the bottle bottoms. Semantic consistency also enables pose estimation for poorly textured and thin-structured objects.

With the initialized poses, we jointly optimize the neural implicit representation of the object and the poses during the interaction. While joint optimization can refine the poses in many cases, it struggles with frames that have severe occlusions between the hand and the object due to significant deviations in the initial poses. These outlier poses lead to incorrect optimization and hinder pose self-correction. To address this problem, we propose a pose outlier voting strategy to check inter-view consistency with Sampson distance in the local windows of input sequences. The detected outliers are reset with poses propagated from neighboring inliers during optimization.

We evaluate our method on three datasets: HO3D [12], HOD [18], and HOPE [47], which contain objects with diverse geometries and textures interacting with one or both hands. The experiments show that our approach significantly outperforms state-of-the-art methods in terms of reconstruction quality and demonstrate its real-world applicability through in-hand scanning with a mobile phone, as shown in Fig. 1. To summarize our contributions:

- We propose a novel method for hand-held object reconstruction under dynamic interaction with one or both hands using priors from pre-trained 3D generation model;
- We propose a frame-to-model alignment method with semantic consistency to address the appearance discrepancy in shape and texture between the generated 3D priors and the input RGB sequences;
- We propose a pose outlier voting strategy that leverages Sampson distance to check inter-frame pose consistency, ensuring both pose accuracy and reconstruction quality.

2. Related Work

Text-to-3D. Text-to-3D generation creates 3D assets from text prompts. Building on advances in text-to-image synthesis with diffusion models [16], DreamFusion [34] adapts image-based diffusion models [38] for 3D generation. Subsequent work has improved generation quality, training stability, and speed [29, 31, 41, 44]. Another approach [17, 22, 26, 52] uses Large Reconstruction Models to generate 3D structures in a feed-forward manner. Building on these advances, we can generate high-quality meshes from text to guide coarse pose initialization.

Multi-view 3D Reconstruction. Recovering 3D geometry from multi-view images has a long history in computer vision and graphics. Traditional approaches typically utilize Structure-from-Motion (SfM) [39] to estimate camera poses, followed by multi-view stereo (MVS) to generate dense 3D reconstructions using explicit representations [3, 11, 25]. Recently, neural implicit representations [32] have effectively modeled 3D geometry and appearance through volume rendering. NeuS [49] and VolSDF [54] enhance surface quality by replacing density fields with signed distance fields. While these methods achieve high-quality reconstructions, they rely on accurate pose estimates, typically obtained with COLMAP [40]. However, COLMAP assumes static scenes, making it unsuitable for scenarios like ours, where hand-object movement is independent of the background. Recent works [20, 28] address this by jointly optimizing poses and implicit representations with added supervision from correspondences [35, 46] or monocular depth [2], but they are effective only in forward-facing scenarios. In contrast, we approach this problem by proposing effective initial poses, a critical factor for successful joint shape-pose optimization [50].

Hand-held Object Reconstruction. Reconstructing 3D manipulated objects is challenging due to occlusions and object diversity. Several methods [5, 53] rely on ground-truth CAD models, simplifying the task to 6DoF pose estimation. Other learning-based approaches attempt to directly infer representations (*e.g*., mesh [6, 14], SDF [7, 23, 56]) from a single RGB(D) image. Unfortunately, these methods suffer from poor generalization due to limited 3D hand-object data, and the reconstructed results often lack detail. CHORD [27] leverages categorical object priors for improvement but remains constrained to narrow category ranges. Recent advancements [9, 18, 21, 42, 57, 58] use

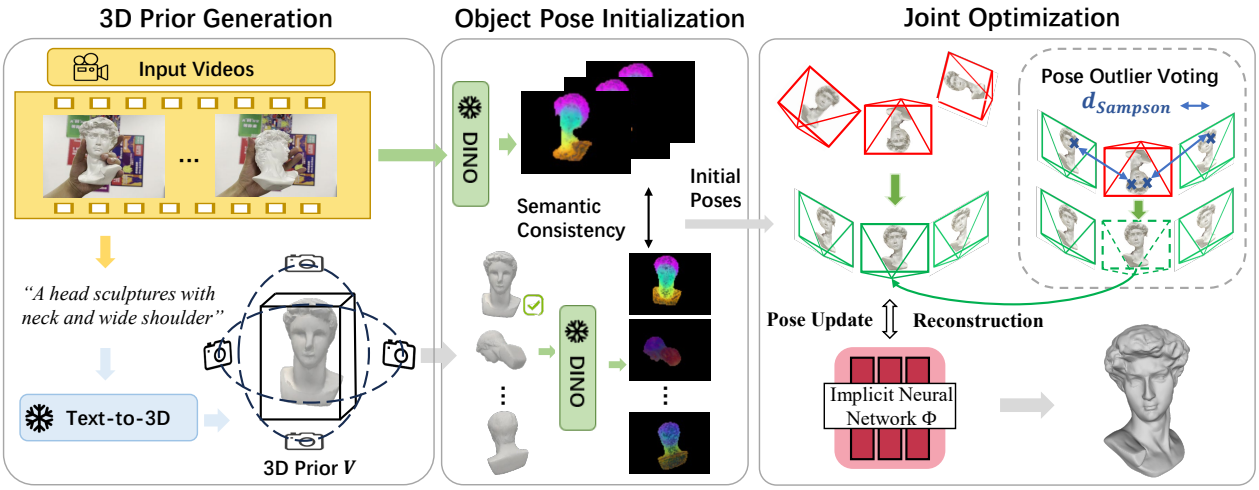


Figure 2. **Overview of our framework.** We first prompt ChatGPT to describe the hand-held object and generate a 3D textured mesh based on this textual description. Despite appearance discrepancies, we leverage semantic consistency to initialize the object pose by aligning the generated prior with the input images in the DINO feature space. We then use the coarse poses to learn an implicit object representation while simultaneously refining the poses. Finally, we introduce a pose outlier voting strategy to correct significantly erroneous poses.

volume rendering to aggregate observations across multiple rigid poses, improving generalization without 3D supervision. However, they still rely on COLMAP for pose initialization or assume a firm grasp, replacing object motion with hand motion, which hinders reconstruction of texture-less or thin objects and limits them to single-hand interactions. Recently, Swamy et al. [43] enhances COLMAP with a learning-based rigid transformation estimation framework. BundleSDF [51] can estimate object pose while simultaneously reconstructing surfaces but requires depth information. In contrast, our method reconstructs high-quality objects during dynamic interactions involving one or two hands from just RGB sequences. Most related to our work, Hampali et al. [13] divides video sequences into short segments to incrementally estimate shape and pose, but this introduces pose errors at segment boundaries, degrading reconstruction quality. Differently, our method jointly produces accurate pose and shape estimations.

3. Methods

An overview of our method is illustrated in Fig. 2. Given a monocular RGB video sequence $\{I_k\}_{k=1,\dots,N}$ of a hand (or both hands) interacting with a rigid object, our goal is to reconstruct the target 3D object shape. The key idea is to use generated 3D meshes V as the priors to obtain the initial object poses $\hat{\xi}$ in the input frames for the reconstruction. More specifically, we first generate V based on the text description. As shown in Fig. 2, although there may be discrepancy in shape and texture between the generated 3D mesh and input and directly aligning them under the discrepancy is challenging, the generated mesh remains semantically similar and can help to constrain the pose space. To this end, we estimate object pose $\hat{\xi}_k$ for the k_{th} frame by aligning the input image I_k with renderings in the foundational semantic-feature space $\mathcal{F}_{DINO}$ (Sec. 3.2). Subsequently, we use initial poses $\{\hat{\xi}_k\}_{k=1,\dots,N}$ to jointly optimize the SDF-based object implicit representation Φ and poses (Sec. 3.3). To reduce the impact of poses with significant errors, we introduce a pose outlier voting strategy that detects and reinitializes these poses (Sec. 3.4), ultimately yielding higher-quality geometry and more accurate poses.

3.1. Preliminaries: Object Representation

We briefly introduce object representation using the Signed Distance Function (SDF) optimized via volume rendering. Following NeuS [49], we represent the geometry and appearance using two MLP networks, a geometry network $\Phi_g : \mathbb{R}^3 \rightarrow \mathbb{R}$ and a color network $\Phi_c : \mathbb{R}^3 \times \mathbb{S}^2 \rightarrow \mathbb{R}^3$. Given a 3D point x, the geometry network outputs the SDF value $\Phi_g(x)$, while the color network takes x and view direction d as inputs to produce the color $\Phi_c(x,\ d)$.

For each pixel, we sample a set of points along the camera ray r, denoted as $\{p(t_i) = o + t_i d \mid t_i \in [t_n,\ t_f]\}$, where $p(t_i)$ are the sampled points, o is the camera position and t_n, t_f denote the bounds of the sample ray. Then we can get the rendered color as:

$$\hat{C}(r) = \sum_{i}^{i-1} \prod_{j=1} (1 - \alpha_j)\alpha_i \Phi_c(p(t_i),\ d), \qquad (1)$$

where $\alpha_i = 1 - \exp(-\int_{t_i}^{t_{i+1}} \rho(t)dt)$ denotes the opacity values. $\rho(t)$ is the opacity density transferred from SDF as defined in [49], and the approximate surface depth [10] can

be interpolated from predicted SDF using:

$$\frac{\Phi_g(p(t_i))t_{i+1} - \Phi_g(p(t_{i+1}))t_i}{\Phi_g(p(t_i)) - \Phi_g(p(t_{i+1}))}, \tag{2}$$

where $\Phi_g(p(t_i))\Phi_g(p(t_{i+1})) < 0$. The object representation Φ can be optimized with photometric loss:

$$\mathcal{L}_{color} = \sum_r \|\hat{C}(r) - C(r)\|, \tag{3}$$

measuring the difference between the rendered color $\hat{C}(r)$ and the input color $C(r)$. The Eikonal loss and mask loss are applied to regularize the SDF, as proposed in NeuS:

$$\mathcal{L}_{eik} = (||\nabla\Phi_g||_2 - 1)^2, \tag{4}$$

$$\mathcal{L}_{mask} = \sum_r BCE(M(r),\ \hat{W}(r)), \tag{5}$$

where $\hat{W}(r) = \sum_i \prod_{j=1}^{i-1}(1-\alpha_j)\alpha_i$ represents the accumulated weight along the ray, and $M(r)$ indicates the corresponding object mask value.

3.2. Pose Initialization via Generated 3D Prior and Semantic Consistency

Generating a textured mesh V as the prior for pose initialization takes three steps: (1) We prompt ChatGPT [1] to describe the shape and appearance of the hand-held object using the first frame I_1 [1]; (2) The response of ChatGPT serves as the input prompt for the Text-to-3D model, with Genie [45] as the generator, producing high-quality 3D textured meshes; (3) We generate N_V meshes and compute the 3D-image similarity between each mesh and all frames using OpenShape [30], selecting the mesh with the highest score as V. This approach enables fully automated mesh generation that is consistent with the input sequence.

We then initialize the object pose in each frame of the input sequence. Given an RGB frame I_k and the generated 3D textured mesh V, our goal is to estimate the object pose $\hat{\xi}_k$ for the frame, which transforms the object from its canonical space to the camera space. $\hat{\xi}_k$ includes 3D rotation $R_k \in SO(3)$ and translation $T_k \in \mathbb{R}^3$. Previous works [5, 15] render 2D mask M_{render} from the 3D mesh and compare it with the input object mask M for optimization:

$$\mathcal{L}_{proj} = \|M_{render} - M\|. \tag{6}$$

However, these methods are sensitive to shape discrepancies and can lead to ambiguities, since multiple object poses may produce similar 2D masks, as shown in Fig. 3.

To address these issues, we additionally align rendered images to input images using semantic feature similarity,

[1]We can also use additional frames, repeat the following steps for each frame, and select the generated mesh with the highest score.

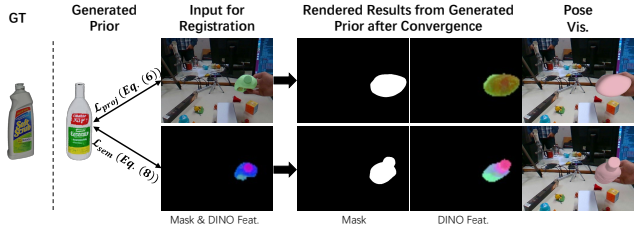


Figure 3. **Top:** Due to shape discrepancy, relying solely on $\mathcal{L}_{proj}$ cause errors (upside down). **Bottom:** Incorporating semantic consistency ensures accurate alignment of 3D model with the input image, despite variations in shape and texture.

which offers greater tolerance to variations in both shape and texture. Specifically, pre-trained foundational models such as DINO [33] have demonstrated their effectiveness as semantic feature extractors [61]. The pixel-wise semantic similarity between two images, I_i and I_j, can be measured as:

$$\mathcal{F}_{cos}(I_i, I_j) = \frac{< \mathcal{F}_{DINO}(I_i),\ \mathcal{F}_{DINO}(I_j) >}{\|\mathcal{F}_{DINO}(I_i)\|_2 \|\mathcal{F}_{DINO}(I_j)\|_2}, \tag{7}$$

where $\mathcal{F}_{DINO}$ represents the feature maps extracted by DINO. We then define the semantic difference between the rendered image $DR(V,\ \hat{\xi}_k)$ and the input image I_k as:

$$\mathcal{L}_{sem} = \|1 - \mathcal{F}_{cos}(I_k,\ DR(V,\ \hat{\xi}_k))\|, \tag{8}$$

where $DR(\cdot)$ represents the differentiable texture rendering operation. We focus only on the object pixels. Combining the $\mathcal{L}_{proj}$ and $\mathcal{L}_{sem}$, we estimate the object pose $\hat{\xi}_k$ by minimizing the energy function via render-and-compare:

$$\min_{\hat{\xi}_k} \lambda_{proj}\mathcal{L}_{proj} + \lambda_{sem}\mathcal{L}_{sem}. \tag{9}$$

The energy is optimized for each frame individually. As objects move continuously in the sequences, we utilize an additional term to the above energy and perform optimization for all the input frames to enforce temporal smoothness:

$$\sum_{k=1}^{N-1} \|(R_{k+1} - R_k)V + (T_{k+1} - T_k)\|. \tag{10}$$

3.3. Joint Optimization with Geometric Constraints

After obtaining the coarse poses $\hat{\xi}$, we convert the video frames into calibrated multi-view images in object-centric coordinates for implicit representation optimization. However, $\hat{\xi}$ are imperfect, which can directly degrade reconstruction quality. BARF [28] demonstrates that jointly optimizing the implicit representation and camera poses mitigates pose noise. Building on this, we treat camera poses as optimizable parameters for joint shape-pose optimization.

Additionally, we incorporate two types of on-surface geometry cues, multi-view correspondence [46] and monocular normal prior [59] as constraints in the optimization for further improvement. Specifically, given the predicted monocular normal $\bar{N}$, the normal consistency loss is formulated as:

$$\mathcal{L}_{normal} = \sum_{r} \|1 - \hat{N}(r)^T \bar{N}(r)\|, \tag{11}$$

where the rendered normal $\hat{N}(r)$ can be computed similarly to color using Eq. (1). Furthermore, for an image I_i and its randomly selected neighboring frame I_j, we extract dense correspondences using DKM [8], resulting in pairs of matching pixels $p_i \in I_i$ and $p_j \in I_j$, along with a correspondence confidence w_p. We define the multi-view correspondence objective as:

$$\mathcal{L}_{corres} = \sum_{p} w_p \|p_j - \pi(\xi_j \xi_i^{-1}\, \pi^{-1}(p_i,\ \hat{z}(p_i;\Phi)))\|, \tag{12}$$

where $\pi(\cdot)$ represents the camera projection operator, and $\hat{z}(\cdot;\Phi)$ is the predicted surface depth computed from the implicit representation Φ using Eq. (2). The overall geometry constraints are then defined as:

$$\mathcal{L}_{geo} = \lambda_{normal}\mathcal{L}_{normal} + \lambda_{corres}\mathcal{L}_{corres}. \tag{13}$$

Incorporating $\mathcal{L}_{geo}$ facilitates joint optimization, improving both pose accuracy and reconstruction quality. The total loss for training the object's implicit representation is:

$$\mathcal{L} = \mathcal{L}_{color} + \lambda_{eik}\mathcal{L}_{eik} + \lambda_{mask}\mathcal{L}_{mask} + \mathcal{L}_{geo}, \tag{14}$$

where $\mathcal{L}_{color}$, $\mathcal{L}_{eik}$, and $\mathcal{L}_{mask}$ are defined in Sec. 3.1.

3.4. Pose Outlier Voting with Sampson Distance

Although the poses from initialization can be refined by joint optimization in many cases, severe occlusions between the hand and object can lead to significant deviations in the initial poses and these pose outliers result in erroneous joint optimization and hinder the refinement. To address this issue, we propose a pose outlier voting strategy to detect and correct the poses.

One solution to find the outliers is to use per-frame rendering quality PSNR [19] to indicate inaccuracies in the corresponding camera pose ξ_k. However, we observe that inaccurate camera poses can also affect neighboring viewpoints due to the implicit network learning incorrect shape and texture in these regions, which can lead to erroneous assessments when using the PSNR of neighboring rendered images.

Instead, we additionally utilize the previously obtained correspondence cues as a metric to help detect erroneous poses. Given pairs of 2D matched points $p_i \in I_i$ and $p_j \in I_j$, along with the fundamental matrix F obtained from the current camera poses ξ_i and ξ_j, we compute the Sampson distance:

$$d_{Sampson} = \frac{|p_i^T F p_j|}{|(p_i F)^1 + (p_i F)^2 + (F p_j)^1 + (F p_j)^2|}, \tag{15}$$

where $(p_i F)^1$ and $(p_i F)^2$ refer to the first and second components of the respective vectors. Unlike $\mathcal{L}_{corres}$ (Eq. (12)), $d_{Sampson}$ relies solely on current camera poses and 2D correspondences, bypassing depth inferred from the network to eliminate neighbor bias. Using this distance metric, we define the Sampson inconsistency rate β as the ratio of pairs where $d_{Sampson}$ exceeds a threshold, relative to the total number of pairs.

We then combine the Sampson inconsistency rate β and rendering quality PSNR as criteria to detect pose outliers, which are subsequently reset with poses propagated from neighboring inliers. Specifically, given a local window of $2K$ neighboring frames centering on frame k, we compute PSNR_k and the average of Sampson inconsistency rate $mean(\{\beta_k\})$ between frame k and the other frames in the window. If PSNR_k is less than λ standard deviations below the mean rendering quality in the window: $\text{PSNR}_k \leq mean(\{\text{PSNR}_i\}_{i=k-K}^{k+K}) - \lambda \cdot std(\{\text{PSNR}_i\}_{i=k-K}^{k+K})$, and its average Sampson inconsistency rate $mean(\{\beta_k\})$ exceeds a certain threshold, the current pose ξ_k is voted for low quality. We then use spherical interpolation to reinitialize low-quality poses with nearby high-quality poses, leveraging the smooth nature of camera motion in videos. This pose outlier voting process helps prevent the shape-pose joint optimization from converging to sub-optimal solutions and supports self-correction.

4. Experiments

4.1. Implementation Details

We use SAM-v2 [37] and StableNormal [55] to generate object masks and monocular normals for each frame. The number of generated meshes, N_V, is set to 12, and render the textured meshes using PyTorch3D [36]. Semantic features are extracted with DINO-ViT-B/14. We adopt the same architecture for the implicit network as NeuS [49]. During training, we employ the Adam optimizer [24] with a learning rate of $5e-4$ and a cosine decay schedule with $\alpha = 0.05$. We sample 1024 rays per batch over a total of 120k iterations. At 20k iterations, we detect erroneous poses and reinitialize the pose and geometry MLPs. Training for 100 frames takes about 1 hour for pose initialization and 8 hours for reconstruction on an NVIDIA RTX 3090 GPU. Our implementation is based on PyTorch.

	HO3D						HOD
	CD (cm^2)↓	F5 (%)↑	F10(%)↑	RPE$_t$(cm)↓	RPE$_r$(°) ↓	ATE(m)↓	CD↓
IHOI [56]	4.78	53.6	77.4	-	-	-	7.28
HOLD [9]	1.82	63.6	85.1	4.143	5.974	0.265	1.27
Ours	**0.51**	**78.2**	**95.4**	**2.129**	**2.962**	**0.072**	**0.44**

Table 1. **Comparison with SOTA hand-object reconstruction methods on HO3D and HOD datasets.** We report the object reconstruction quality and pose accuracy. Chamfer Distance (CD) for HOD dataset is reported in unit size.

	CD (cm^2)↓	F5 (%)↑	F10 (%)↑
Hampali et al. [13]	1.4	57.4	79.9
Ours	**0.6**	**76.0**	**94.4**

Table 2. **Comparison with SOTA in-hand scanning method on the same HO3D sequence as Hampali et al. [13].**

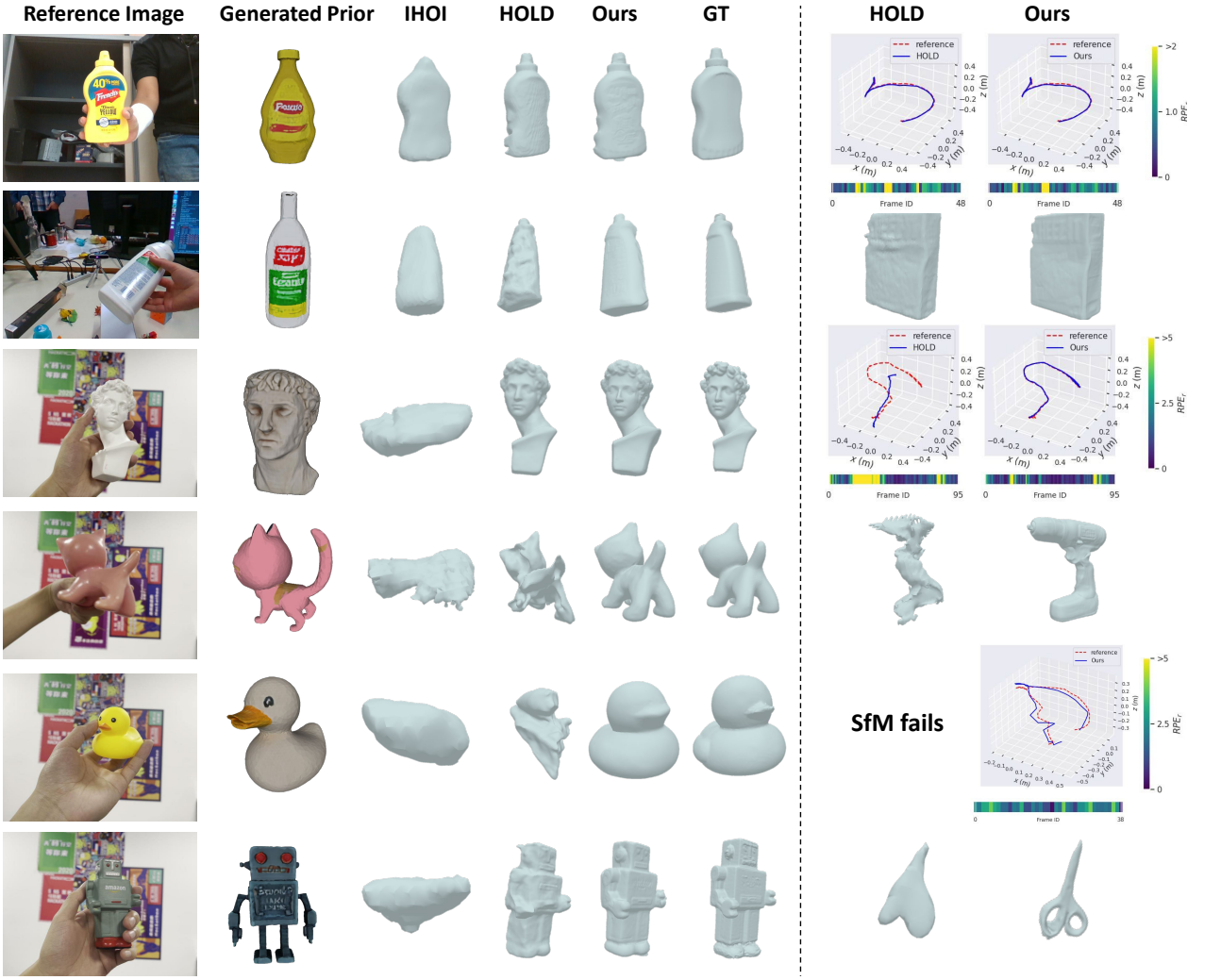


Figure 4. **Qualitative Comparison with SOTA.** We present reconstructed meshes from our method and baselines on HO3D [12] and HOD [18]. Pose trajectories on HO3D are also visualized, with ground-truth poses in red and predicted poses in blue. The color bar at the bottom indicates the relative rotation error, with lighter colors representing larger errors. Additionally, we show the generated prior used by our method. We yield more accurate results than other approaches, particularly for texture-less objects. Zoom in for details.

4.2. Datasets

We evaluate our method using three real-world datasets, each featuring different types of objects and interactions. **HO3D [12]:** This dataset consists of RGB videos of a hand interacting with YCB objects [4] with 3D annotations of the object and ground truth poses for each frame. We use the latest version, HO3D_v3, and select two sequences per object for evaluation, one of which matches [13], resulting in a total of 20 sequences.

HOD [18]: This dataset includes videos of hand-held objects in motion, recorded by a static RGB camera with ground-truth meshes. Compared to HO3D, HOD features

objects with more complex shapes and challenging appearances, making it better suited for evaluating hand-held object reconstruction. We use 7 sequences to assess object reconstruction quality.

HOPE [47]: This dataset contains videos of two hands rotating grocery objects. We use the latest version, HOPEv2, and select 8 sequences to evaluate our method for reconstructing objects manipulated by both hands.

4.3. Metrics and Baselines

Metrics. Following previous works [9, 18, 56], we use the **Chamfer Distance** (CD) to evaluate object reconstruction quality and the **F-score** (in percentage) to assess local shape details. Specifically, we apply ICP to register the reconstructed mesh with the ground-truth. For ground-truth meshes with known scale (*e.g.*, HO3D, HOPE), we compute the Chamfer Distance in squared centimeters and the F-score at 5mm (F5) and 10mm (F10). For HOD, where the scale is unknown, meshes are normalized to unit size, and only the Chamfer Distance is reported. Additionally, we evaluate pose accuracy using standard visual odometry metrics [62], including Absolute Trajectory Error (**ATE**) and Relative Pose Error (**RPE**). ATE measures the difference between estimated and ground truth camera positions, while RPE quantifies relative rotation (RPE_r) and translation (RPE_t) errors between image pairs. We align the predicted trajectory with ground truth via Umeyama alignment [48], reporting RPE_r in degrees, RPE_t in centimeters, and ATE in meters.

Baselines. We compare our method with several existing approaches, including **HOLD** [9], the state-of-the-art for reconstructing hand-object interaction from monocular RGB video, which relies on SfM to initialize the object pose. As the method fails to reconstruct the banana and scissors in HO3D, HOLD evaluates reconstruction quality by adding noise to the ground truth poses for initialization on these two objects. The evaluation of HOLD in Tab. 1 adopts their settings. When calculating the pose trajectory metric, we exclude these parts. **IHOI** [56], a learning-based method for monocular-image in-hand object reconstruction, trained on sequences from HO3D and other datasets. We evaluate IHOI on each frame of the sequence and report the average results. **Hampali et al. [13]** shares the same settings as ours. Without available code, we follow HOLD and compare against their released 3D models reconstructed by their method on the same HO3D sequences, excluding the banana and scissors due to their known failure cases. In two-hand scenarios, where no available prior work has addressed this challenge, we compare with BARF [28]. For a fair comparison, we provide initialization poses from COLMAP, similar to HOLD but without hand refinement, referring to this baseline as **COLMAP-BARF**.

	CD (cm^2)↓	F5 (%)↑	F10(%)↑
COLMAP-BARF	0.96	62.5	79.8
Ours	**0.29**	**85.4**	**96.9**

Table 3. **Comparison with baseline on two-hand interaction object reconstruction.**



Figure 5. **Qualitative results of two-handed object reconstruction.**

4.4. State-of-the-art Comparsion

Single-handed object reconstruction. We visualize the reconstructed object meshes and estimated pose trajectories in Fig. 4. The learning-based method IHOI can predict coarse shapes on seen data but lacks fine details. It also fails to reconstruct unseen objects from the HOD dataset, generating invalid meshes. HOLD can accurately reconstruct geometry for objects with rich features, but its performance is significantly limited by its object pose initialization via SfM, as shown in the trajectories. For example, SfM gets confused by the similar appearance of the two sides of the drill, resulting in completely incorrect pose initialization. Additionally, SfM struggles to initialize correct poses for texture-less or thin objects, leading to poor-quality reconstruction results. In contrast, by incorporating the generated prior, our method achieves superior geometry quality and accurate pose trajectories, even for challenging objects like scissors, the cat, and the rubber duck.

This is consistent with quantitative results in Tab. 1. Our method significantly outperforms comparison methods on reconstruction metrics on both datasets, particularly on the HOD with more challenging objects. Moreover, our method also surpasses HOLD in pose accuracy, as its performance is constrained by SfM-based initialization. Tab. 2 compares our results with Hampali et al. [13] using the same HO3D sequences, showing a clear improvement in reconstruction quality. Detailed qualitative comparisons are provided in the supplementary material.

Two-handed object reconstruction. In two-hand interaction settings, our method consistently outperforms the baseline, as shown in Tab. 3. The HOPE dataset mainly contains textured objects, reducing the difficulty of object pose initialization with COLMAP and joint shape-pose optimization with BARF. However, as illustrated in Fig. 5, its results still contain numerous artifacts, whereas our reconstructed meshes demonstrate higher fidelity.

	Pose initialization				After joint optimization			
	Prior-CD (cm^2)↓	RPE$_t$(cm)↓	RPE$_r$(°) ↓	ATE(m)↓	CD (cm^2)↓	RPE$_t$(cm)↓	RPE$_r$(°) ↓	ATE(m)↓
GT	0.0	**2.019**	**3.049**	**0.077**	**0.43**	**1.587**	**2.328**	**0.043**
Gen$_{Genie}$	1.05	3.629	4.738	0.115	0.51	2.129	2.962	0.072
Gen$_{Shap-e}$	1.67	4.178	5.519	0.153	1.11	2.931	4.108	0.122

Table 4. **The influence of different priors evaluated on HO3D.** Left: pose initialization results. Prior-CD indicates the Chamfer Distance between generated priors and ground truth. Right: object reconstruction quality and pose accuracy after joint optimization.

Pose Init.	$\mathcal{L}_{geo}$	Re-Init.	CD (cm^2)↓	RPE$_t$(cm)↓	RPE$_r$(°) ↓	ATE(m)↓
			-	7.948	10.250	0.385
w/o $\mathcal{L}_{sem}$			4.17	5.686	8.277	0.246
✓			0.65	2.772	3.852	0.093
✓	✓		0.62	2.443	3.347	0.084
✓	✓	PSNR	0.58	2.206	3.052	0.075
✓	✓	✓	**0.51**	**2.129**	**2.962**	**0.072**

Table 5. **Ablation study of joint shape-pose optimization on HO3D.** w/o $\mathcal{L}_{sem}$ represents pose initialization relying solely on masks with generated priors. PSNR denotes the naive pose re-initialization strategy without $d_{Sampson}$.

4.5. Ablation Studies

The influence of different object priors. We show the influence of generated priors on pose initialization and the joint optimization results in Tab. 4. We utilize priors from two text-to-3D methods (Shap-e [22] and Genie [45]) and compute the Chamfer Distance between the generated priors and ground-truth models (**Prior-CD**), where a lower value indicates better similarity. We find that our method is robust to priors of varying quality and $\mathcal{L}_{sem}$ is crucial for pose initialization. GT refers to using ground-truth templates for pose initialization, which sets the upper bound for our method. Although Shap-e [22] (Gen$_{Shap-e}$) generates lower-quality priors compared to Genie [45] (Gen$_{Genie}$), leading to larger pose initialization errors, it still provides a reasonable initialization. This is further evident in the final object reconstruction quality, where joint optimization with Shap-e initialization consistently outperforms the SOTA methods on HO3D (Tab. 1). Additionally, the object reconstruction quality with Genie initialization is nearly identical to that with GT initialization. These results demonstrate the effectiveness of our pose initialization and the robustness of our system to different generated priors.

The effectiveness of components in joint shape-pose optimization. We perform ablation studies by incrementally incorporating each component into the joint optimization process. As shown in Tab. 5, all the proposed techniques prove essential. The top row initializes all poses from the identity matrix and simply performs joint optimization, which is equivalent to BARF. This approach fails to learn both the correct pose and the object shape. In contrast, providing our pose initialization significantly improves the reconstruction quality. The table shows that $\mathcal{L}_{sem}$ is crucial for initialization, as omitting $\mathcal{L}_{sem}$ in pose initialization severely degrades the overall reconstruction. Adding

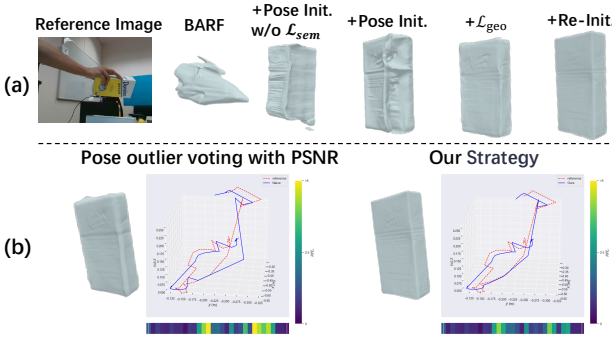


Figure 6. **Ablation study.** (a) Each proposed component effectively improves reconstruction quality. (b) Our voting strategy further reduces the impact of large pose errors and enhances reconstruction quality compared to relying solely on PSNR.

$\mathcal{L}_{geo}$ further improves overall pose accuracy and mesh quality, although some artifacts persist. Visualizations in Fig. 6 demonstrate that our voting strategy further refines surface quality. We also compare to the strategy that solely based on rendering quality. As indicated by the trajectory, our strategy results in smaller pose errors and achieves lower Chamfer Distance (CD) values, highlighting its effectiveness.

5. Conclusion

In this paper, we present a method to reconstruct high-quality hand-held objects from monocular RGB video with dynamic interactions. The key insights include incorporating generated 3D priors and enforcing semantic consistency for coarse pose initialization, which significantly supports the optimization of the object's implicit representation. We also introduce a pose outlier voting strategy to further enhance both reconstruction quality and pose accuracy. Our method consistently achieves state-of-the-art results across multiple datasets.

Limitations and discussion: Despite promising results, our method has certain limitations. Its effectiveness depends on the quality of generated priors, which may not perform well with unique objects that current generators struggle to represent accurately. Additionally, DINO may produce nearly identical features for different sides of an object, leading to incorrect pose estimations in some cases. Advances in semantic correspondences [60] may help mitigate this issue. Nevertheless, we believe this work provides valuable insights for reconstructing objects from unposed multi-view images.

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